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A mechanistic model for pipeline steel corrosion in supercritical CO_2–SO_2–O_2–H_2O environments

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ABSTRACT

A mechanistic model was established to predict uniform corrosion rate and investigate the corrosion mechanisms of pipeline steel in supercritical CO_2–SO_2–O_2–H_2O environments. A six-region division (SIWDES: Supercritical CO_2, Interface, Water film, Deposition, Electrodic, and Solid) was applied to mathematically describe the model. The modified three-characteristic-parameter correlation model was used to calculate the ion activity coefficients, which calculates the activity coefficients of ions in thin water film with high ionic strength. The model can reasonably predict the corrosion rate of pipeline steel for the primary variables, determine the concentration distribution of each component in the water and product films, and also reflect the impact of corrosion product film on corrosion rate. A comparative analysis between the model and the experimental results showed that the model reasonably predicted the effects of the main factors on corrosion rate.

1. Introduction

Prediction of corrosion rate based on the models of physicochemical processes is one of the hotspots of research in the fields of corrosion science and technology because it can assist engineers in making decisions related to corrosion issues. Prediction of corrosion rate is extremely important in oil and gas industries, and many scholars have carried out a series of detailed studies regarding this topic [1–14].

Carbon capture and storage (CCS) is one of the alternative ways to mitigate climate change problems. CO_2 captured from power plants is primarily transported by pipeline or ship to storage sites [15]. CO_2 containing impurities reportedly exerts a corrosive effect on the inner wall of the pipeline, but this issue remains controversial [16]. Some experimental studies regarding the effect of SO_2–O_2–H_2O impurities on the corrosion behavior of pipeline steels have been carried out [17–28]. However, to date, there is still lacking of a mechanistic model for pipeline steel corrosion in supercritical CO_2 environments, which is important to further understand corrosion mechanisms and to better predict uniform corrosion rate. *Less dense than liquid and more dense than gas.*

CO_2 in the pipeline is either in supercritical or liquid state, and the pressure range is generally between 8.6 and 15.3 MPa for the current CO_2 pipelines [29]. In CO_2 mixtures, CO_2 acts as a solvent, whereas other impurities act as solutes. When the relative humidity of CO_2 is high, a thin water film or water droplet forms on the inner wall of the pipeline, which can lead to electrochemical corrosion. The reactants need to pass the interface between the supercritical CO_2 and thin water film to participate in the corrosion reactions. Corrosion with thin water film always contains an electrolyte solution with high ionic strength.

Although a corrosion model of steel in supercritical CO_2 mixtures is still lacking, some corrosion models such as steel corrosion model in CO_2 water solution and atmospheric corrosion model provided the necessary baseline for this study. The earliest CO_2 corrosion model for the oil and gas pipeline was proposed by de Waard and Milliams [1]. This model has been improved and gradually developed [13]. In recent years, many scholars have proposed various comprehensive and accurate aqueous solution CO_2 corrosion models, such as those proposed by Nesic [3,5], Nordsveen [7], and Song [9,11,12]. Nordsveen's model [7] considered the diffusion process of reactants and products in the product scales, wherein the product scale thickness should be given. Nesic's model [3] involved the effect of product scale growth on corrosion rate. Song's model [9,11,12] studied the cases under aerobic, anaerobic, and cathodic protection conditions.

The situation of steel corrosion in supercritical CO_2 mixtures highly resembles corrosion in atmospheric environments [17]. The Gas, the Interface between gas and liquid, the Liquid, the Deposition layer, the Electrodic region near the surface, and the Solid (GILDES) six-regime model proposed by Graedel [30] is one of the most significant models of corrosion under thin water film conditions in atmospheric environments. The concept of dividing the corrosion

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Nomenclature

A	reaction zone area, m^2
A_ϕ	Debye–Hückel equation constant
c_m	mass concentration, kg/kg
c_j	mole concentration for species j, mol/L
CR	corrosion rate, mm/a
d_f	product film thickness, m
d_w	water film thickness, m
d_i	total thickness of water film and product film, m
d_J	diameter of the blade, m
D	diffusion coefficient, m^2/s
D_m	molecular diffusion coefficient, m^2/s
D_E	turbulent diffusion coefficient, m^2/s
D_e	dielectric constant
e	electron charge, C
EW	equivalent mass, kg
h	space interval, m
$H_{s,e}$	effective Henry's constant for SO_2, mol/(L atm)
H_j	Henry's constant for species j, mol/(L atm)
H_{j0}	Henry's constant for species j (ionic strength is zero), mol/(L atm)
i_{H^+}	hydrogen electrode current density, A/m^2
i_{O_2}	oxygen electrode current density, A/m^2
i_{Fe}	anode current density, A/m^2
I	ionic strength, mol/kg
k	rate constant, s^{-1}
k_B	Boltzmann constant, J/K
K_c	corrosion rate equation coefficient
K_{su}	first ionization equilibrium constant of SO_2
K_{bi}	second ionization equilibrium constant of SO_2
K_{sp}	solubility product constant
K_w	ionization constant of water, mol^2/L^2
L	Avogadro's constant
m	molality, mol/kg
$m(S)$	sum of molality for all components, mol/kg
mp_{SO_2}	SO_2 mole percent
mp_{O_2}	O_2 mole percent
M	molar mass, kg/mol
n	mole, mol
N	diffusion flux, mol/(m^2 s)
p_j	partial pressure of species j, atm
P	total pressure, MPa
R	chemical reaction rate, mol/(m^3 s)
R_G	precipitation rate, kg/(m^2 s)
RH	relative humidity
RPM	stirring speed of the autoclave, rpm
S	solubility, kg/kg
S_n	mole solubility, mol/L
S_S	super saturation
t	time, s
T	temperature, K
U	Velocity, m/s
ν_+	chemical equivalent coefficient of cation
ν_-	chemical equivalent coefficient of anion
x	x coordinate
z	component charge number
ρ	density, kg/m^3
ν	kinematic viscosity, m^2/s
Δc	mass concentration difference, kg/kg
β	salt-out effect coefficient, kg/(L atm)
γ	activity coefficient
τ	time step, s

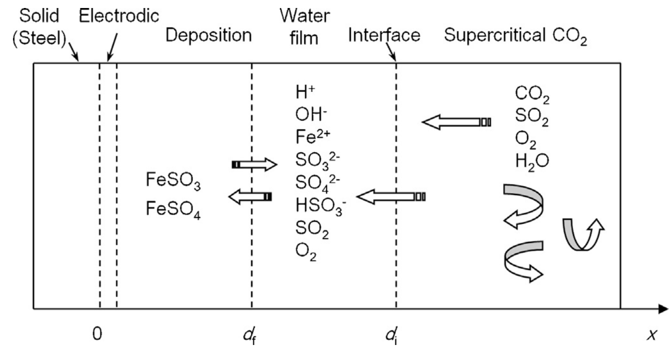

Fig. 1. Schematic diagram of the six regions that make up the model.

process into six layers as GILDES model was adopted in the current model.

Based on Nordsveen's [7] and Nesic's [3] models and GILDES model's [30] method of treating corrosion under thin water film conditions, a mechanistic model for pipeline steel corrosion in supercritical CO_2–SO_2–O_2–H_2O environments was established. Similar approaches were involved and modifications have also been made to solve the new problems encountered with the characteristics of high ionic strength, high pressure, and supercritical fluid environments. The model calculation results are compared with the previous experimental results to verify the reliability of the model.

Various studies have focused on corrosion issues in CO_2 injection and storage processes [31–34]. However, the present study mainly focused on investigating the laboratory experimental conditions related with the corrosion problems of CO_2 transport with SO_2–O_2–H_2O impurities. For further industrial applications, the methods used in top-of-the-line corrosion modeling [10,14] can be referred to in order to properly simulate the industrial environments.

2. Model description

2.1. Variables and assumptions

The model mainly investigates the effects of the following seven variables on corrosion rate: temperature (293–366 K), pressure (8–20 MPa), SO_2 mole percent (0.05%–2.5%), O_2 mole percent (0–2.5%), relative humidity (0–100%), velocity (0–3 m/s), and time.

The whole corrosion process was divided into six layers, as shown in Fig. 1. The water film thickness is as follows:

$$d_w = d_i - d_f \tag{1}$$

To simplify the problems, the following assumptions are reasonably made in this study:

(1) Similar to the models of Nesic [6] and Song [9], this model ignores the contribution of electromigration effect on mass transfer. Nesic [6] stated that the contribution of electromigration to the overall flux of species is small.
(2) The processes that components diffuse in supercritical CO_2 and pass the interface between CO_2 and water film are not the rate-controlling steps. If these steps were always the rate-controlling steps, the corrosion rate would not change significantly with time. The experiment results showed that the corrosion rate decreased sharply during the initial few hours [24]. Actually, components have large diffusion coefficients in supercritical CO_2. The product film seems to have a critical effect on the corrosion process.
(3) The thin water film on the sample surface under non-saturated conditions is not continuous, and the coverage of water

film is positively correlated with the relative humidity. The experiment under low relative humidity showed that corrosion products were discrete points [22]. The separated water droplets seemingly formed on the sample surface.

(4) The solubility of SO_2 in water is significantly larger than that of CO_2 under the same condition. For example, under 25 °C and 1 bar partial pressure condition, the solubility of SO_2 in water is about 1.4 mol/kg [35], whereas that for CO_2 is only 0.035 mol/kg [36]. SO_2 also has a higher degree of ionization than CO_2. Even comparing for the cases with partial pressures of 0.05 bar SO_2 and 100 bar CO_2 at 25 °C, the H^+ concentration generated from the SO_2 dissolution and ionization is 10 times higher than that from CO_2. Therefore, for the case without extremely low SO_2 partial pressure, this model ignores the effect of CO_2 on the corrosion process.

2.2. Chemical reactions

The model assumed the existence of a thin water film during the corrosion process. The following chemical reactions in the thin water film were considered:

Dissolution of SO_2 in the water film:

$$SO_2(g) + H_2O \rightleftharpoons SO_2 \cdot H_2O \tag{2}$$

SO_2 hydrate ionization equilibrium:

$$SO_2 \cdot H_2O \underset{k_{b,su}}{\overset{k_{f,su}}{\rightleftharpoons}} H^+ + HSO_3^- \tag{3}$$

$$HSO_3^- \underset{k_{b,bi}}{\overset{k_{f,bi}}{\rightleftharpoons}} H^+ + SO_3^{2-} \tag{4}$$

Oxidation of sulfite:

$$2HSO_3^- + O_2 \rightarrow 2H^+ + 2SO_4^{2-} \tag{5}$$

$$2SO_3^{2-} + O_2 \rightarrow 2SO_4^{2-} \tag{6}$$

Formation of ferrous sulfate and ferrous sulfite:

$$Fe^{2+} + SO_4^{2-} \rightarrow FeSO_4 \tag{7}$$

$$Fe^{2+} + SO_3^{2-} \rightarrow FeSO_3 \tag{8}$$

Therefore, eight solution components are considered in this model, namely, H^+, $HSO_3{}^-$, $SO_3{}^{2-}$, $SO_2{\cdot}H_2O$, O_2, Fe^{2+}, $SO_4{}^{2-}$, and OH^-. Other negligible trace components are not involved in the detailed calculation.

2.3. Electrochemical reactions

Anodic reaction:

$$Fe \rightarrow Fe^{2+} + 2e^- \tag{9}$$

Cathodic reactions:

$$2H^+ + 2e^- \rightarrow H_2 \tag{10}$$

$$O_2 + SO_2 + 2e^- \rightarrow SO_4^{2-} \tag{11}$$

The hydrogen evolution reaction was experimentally proven to exist in this system [22,23]. The cathodic reaction of oxygen depolarization was also confirmed because some experimental results showed the presence of oxygen accelerated the corrosion reaction [17,18]. Many references [37–39] showed that the corrosion of carbon steel in SO_2-containing environments contains the cathodic reaction in the form of Eq. (11), so it is also considered in this model.

2.4. Mathematical model

Similar to the GILDES model [30], the physical area of this model was divided into six regions as shown in Fig. 1 (SIWDES: Supercritical CO_2, Interface, Water film, Deposition, Electrodic, and Solid regions). The mathematical descriptions of the physical and chemical processes for these regions are as follows. The solid region was ignored in the current model.

2.4.1. Supercritical CO_2 region

The rotary autoclave was used in the previous experiment, and the autoclave stirring speed can be set in rounds per minute. However, the true flow rate of CO_2 mixture fluid in the autoclave cannot be directly determined. In this study, the experimental data regression expression for the velocity of fluid and the rotation speed of autoclave by Wei and Ji [40] is applied to calculate the average velocity of CO_2 mixture fluid in the autoclave:

$$U = 15.0758\mathrm{RPM}(d_J/d_h)^2 \tag{12}$$

2.4.2. Interface region

Similar to the approach in GILDES model [30], this model uses Henry's law to describe this interface process, indicating that the relationship between the dissolved species concentration in the water film and the species partial pressure can be expressed as

$$c_j = H_j p_j \tag{13}$$

If the dissolved gas composition encounters chemical reactions, such as when the ionization of dissolved SO_2 in water generates H^+, $HSO_3{}^-$, and $SO_3{}^{2-}$, the Henry's constant expression is [41]

$$H_{s,e} = \frac{([SO_3^{2-}] + [HSO_3^-] + [SO_2 \cdot H_2O])}{p_{SO_2}} \tag{14}$$

With the increase in exposure time, the ionic strength of the thin water film kept increasing and ultimately formed a concentrated solution with high ionic strength. The presence of electrolytes reduces the solubility of non-electrolytes as part of the salting-out effect, and the Henry's constant is reduced as follows [30]:

$$H_j = H_{j0} - \beta I \tag{15}$$

This model requires an empirical approach to determine the salting-out effect coefficients.

2.4.3. Water film region

All components diffuses in the water and product films, and the mass transfer equation for various components can be expressed as

$$\frac{\partial(\varepsilon c_j)}{\partial t} = -\frac{\partial}{\partial x}\left(\varepsilon^{1.5} D_j \frac{\partial c_j}{\partial x}\right) + \varepsilon R_j - \left(CR - \frac{\partial d_f}{\partial t}\right)\frac{\partial c_j}{\partial x} \tag{16}$$

The first term on the left of the equation is the diffusion term that ignores the electromigration effect. The second term is the chemical reaction source. The third term is the convective term that considers the moving boundary problem caused by the corrosion and the formation of product film. This equation was firstly proposed by Nesic [3], and the current model added a convective term due to the product film thickening.

The treatment of convection diffusion coefficient is similar to Nordsveen's model [7]. The difference is that Nordsveen's model treated one type of fluid for the turbulent diffusion, whereas the present model considers two different fluids (supercritical CO_2 and water) in the boundary layer.

In case of the two fluids in the boundary layer, this model assumes that the thickness of the pseudo-laminar boundary layer δ

is the half of the total pseudo-laminar boundary layer thickness for either water only or supercritical CO_2 only in the boundary layer:

$$\delta = \frac{\delta_{H_2O} + \delta_{CO_2}}{2} \quad (17)$$

The turbulent diffusion in the product film is also ignored in this model. The turbulent diffusion coefficient can be expressed as [42]

$$D_E = 0.18\nu\left[\frac{x - d_f}{\delta}\right]^3 \quad (18)$$

The total diffusion coefficient can be expressed as [6]

$$D = D_m + D_E \quad (19)$$

The thin water film eventually became a concentrated solution, wherein the species in the water film deviated from their idea properties. The activity is introduced to substitute the concentration to be used in calculation. The ionic strength has significant influence on the activities of species and on the corrosion rates. The ionic strength is involved and defined as

$$I = \frac{1}{2}\sum m_j[z(j)]^2 \quad (20)$$

The usual activity coefficient calculation models include Debye–Hückel equation, Davies equation, Pitzer equation, and three-characteristic-parameter correlation (TCPC) model. The Debye–Hückel equation is as follows [43]:

$$\ln\ \gamma = -A_\phi z^2\sqrt{I} \quad (21)$$

This equation is only suitable for a dilute solution.

Based on Debye–Hückel equation, Davies equation has the following expression form [44]:

$$\ln\ \gamma = -A_\phi z^2\left(\frac{I^{1/2}}{1+I^{1/2}} - 0.3I\right) \quad (22)$$

This equation is only suitable for ionic strength less than 0.1 mol/kg. In cases wherein the ionic strength value is large, the Pitzer equations [45] should be used; however, the calculation of this equation is complicated. The modified TCPC model can simply and accurately calculate the ion activity coefficient, which is shown as follows [46]:

$$\ln\ \gamma_\pm = -|z_+z_-|A_\phi\left[\frac{I^{1/2}}{1+b_1I^{1/2}} + \frac{2}{b_1}\ln(1+b_1I^{1/2})\right] + \frac{S_1}{T}\frac{I^{2n_1}}{\nu_+ + \nu_-} \quad (23)$$

A_ϕ can be calculated as [46]

$$A_\phi = \frac{1}{3}(2\pi L\rho_w)^{1/2}\left(\frac{e^2}{D_e k_B T}\right)^{3/2} \quad (24)$$

A_ϕ is 0.392 at 298.15 K [46], and it has a relation with temperature as

$$A_\phi = A_{\phi,0}\left(\frac{D_{e,0}T_0}{D_e T}\right)^{3/2} \quad (25)$$

The activity coefficient calculation method above is for electrolyte, and for the non-electrolytes, such as SO_2 and O_2, they have different calculation methods. For SO_2, the following equation is used [47]:

$$\log(\gamma_{SO_2}) = \left(\frac{0.0997 - 22.3}{T}\right)I \quad (26)$$

For O_2, the following method is used [48]:

$$\ln(\gamma_{O_2}) = 2m(S)\lambda_{O_2,S} + m(S)_c m(S)_a \zeta_{O_2,S} \quad (27)$$

The other two parameters can be calculated as

$$\lambda_{O_2,S} = \frac{0.4341 - 255.5911}{T} + \frac{45132.321}{T^2} \quad (28)$$

$$\zeta_{O_2,S} = -0.0187 \quad (29)$$

Usually, pipeline steel contains a considerable amount of Mn, which can be corroded into Mn^{2+} ion. This ion has a catalytic oxidation effect on sulfite [49]. The catalytic oxidation rate of sulfite is generally expressed as [50]

$$R = kc^m_{SO_3^{2-}}c^n_{O_2}c^q_{Mn^{2+}} \quad (30)$$

In cases wherein relative humidity is low, the dissolution of the anode is difficult and the anode process becomes the rate-controlling step. In this model, when the relative humidity is below 100%, the parameter water film coverage θ is introduced to show the proportion of specimen surface area covered by the water film. The water film coverage and thickness are determined by the user. The calculation of water condensation rate is not included in this model.

2.4.4. Deposition region

The corrosion products were mainly composed of $FeSO_3$ and $FeSO_4$ crystalline hydrates [19,20]. $FeSO_3$ almost does not dissolve in pure water and dissolves a small amount when SO_2 presents. Although a part of sulfite oxidized to sulfate, $FeSO_3$ product film was also generated at the early stage of the corrosion reaction [19].

The $FeSO_4$ product film did not appear at the early stage of the corrosion reaction because it has a high solubility in water. However, as the corrosion reaction proceeded, more $FeSO_4$ was generated. When the water film reached the super saturation state, $FeSO_4$ was deposited to form the solid product film.

Similar to the GILDES model [30], the product film formation rate can be calculated as follows [51]:

$$R_G = k_G\Delta c^{n_s} \quad (31)$$

Parameter n_s is related to the size and morphology of the crystals, and its value is generally between 1 and 2, usually taken as 1.6 [51]. Δc is the difference of mass concentration for $FeSO_3$ or $FeSO_4$. The following expression can be used to calculate the variables:

$$\Delta c_1 = c_{m1} - S_{FeSO_3} \quad (32)$$

$$\Delta c_2 = c_{m2} - S_{FeSO_3} \quad (33)$$

The concentration difference and super saturation has the following relationship:

$$\Delta c = S(S_S - 1) \quad (34)$$

When multiple solvents have the same ion, direct determination of the mass concentration difference Δc is difficult. Therefore, super saturation S_S is used in the following precipitation rate formula:

$$R_G = k_G[S(S_S - 1)]^n \quad (35)$$

For $FeSO_3$ and $FeSO_4$, S_S can be calculated using the following expressions:

$$S_{S1} = \frac{([Fe^{2+}]\cdot[SO_3^{2-}])}{K_{sp1}} \quad (36)$$

$$S_{S2} = \frac{([Fe^{2+}]\cdot[SO_4^{2-}])}{K_{sp2}} \quad (37)$$

K_{sp} has a relationship with mole solubility S_n (mol/L) as [52]

$$K_{sp} = S_n^2 \quad (38)$$

According to the solubility data of $FeSO_3$ and $FeSO_4$ in water, the solubility product can be determined, including the variation of solubility product with temperature.

The precipitation rates of $FeSO_3$ (containing three crystal water) and $FeSO_4$ (containing four crystal water) are defined as R_{G1} and R_{G2}, respectively. The total amount of precipitated $FeSO_3$ and $FeSO_4$ can be respectively expressed as

$$n_{FeSO_3(s)} = \int_0^t \frac{R_{G1}A}{M_{FeSO_3 \cdot 3H_2O}} dt \tag{39}$$

$$n_{FeSO_4(s)} = \int_0^t \frac{R_{G2}A}{M_{FeSO_4 \cdot 4H_2O}} dt \tag{40}$$

The product film thickness of $FeSO_3$ and $FeSO_4$ can be respectively expressed as

$$d_{f1} = \frac{M_{FeSO_3} n_{FeSO_3(s)}}{\rho_{FeSO_3} A(1-\varepsilon)} \tag{41}$$

$$d_{f2} = \frac{M_{FeSO_4} n_{FeSO_4(s)}}{\rho_{FeSO_4} A(1-\varepsilon)} \tag{42}$$

The average volumetric porosity ε needs to be determined by the user. The total thickness of the product film is

$$d_f = d_{f1} + d_{f2} \tag{43}$$

When the amount of product film was more than a certain amount on the sample surface, there was a part of the corrosion product leaving the sample surface due to the gravity effect. This amount can only be determined empirically in this model.

2.4.5. Electrodic region

For the hydrogen electrode current density i_{H^+} and the anode current density $i_{Fe^{2+}}$, the methods proposed by Nordsveen [7] and Nesic [5,53] are used.

In the oxygen electrode reaction, because the solubility of oxygen in water is low, and the diffusion coefficient is also small, this model supposes that the oxygen electrode reaction is controlled by the diffusion process. The limit diffusion current density of the oxygen electrode reaction is

$$i^d_{lim,O_2} = 2FN_{O_2} \tag{44}$$

where N_{O_2} is the diffusion flux of O_2 to the electrode surface. The oxygen electrode reaction requires the participation of SO_2. Thus, the SO_2 flux should be compared with the O_2 flux to the electrode surface. When the SO_2 flux is less than the O_2 flux, the limit diffusion current density of the oxygen electrode should be

$$i^d_{lim,O_2} = 2FN_{SO_2} \tag{45}$$

where N_{SO_2} is the flux of SO_2 to the electrode surface.

According to the current conservation,

$$i_{Fe} = i_{H^+} + i_{O_2} \tag{46}$$

The corrosion potential E and the anodic current density can be calculated. The relationship between the corrosion rate and the anode current density is as follows [54]:

$$CR = K_c \frac{i_{Fe} EW}{\rho} \tag{47}$$

3. Model solution

3.1. Initial and boundary conditions

The dissolution and ionization equilibrium of SO_2 are rapid processes. The equilibrium state of SO_2 and O_2 in water is the initial condition. The three ionization equilibriums in SO_2 solution are as follows:

$$K_w = \frac{[H^+][OH^-]}{[H_2O]} \tag{48}$$

$$K_{su} = \frac{[H^+][HSO_3^-]}{[SO_2 \cdot H_2O]} \tag{49}$$

$$K_{bi} = \frac{[H^+][SO_3^{2-}]}{[HSO_3^-]} \tag{50}$$

According to the charge conservation,

$$[H^+] = [HSO_3^-] + 2[SO_3^{2-}] + [OH^-] \tag{51}$$

According to the mass conservation,

$$[S(IV)] = [SO_2 \cdot H_2O] + \left[HSO_3^-\right] + \left[SO_3^{2-}\right] \quad -14 \tag{52}$$

The five equations above can obtain $1 \leftarrow 1_0$

$$A_4[H^+]^4 + A_3[H^+]^3 + A_2[H^+]^2 + A_1[H^+] + A_0 = 0 \tag{53}$$

The largest real root of Eq. (53) is the initial concentration of H^+, and the initial concentration of the remaining components can also be obtained, where

$$A_0 = -K_w \tag{54}$$

$$A_1 = -\left(\frac{K_w}{K_{bi}} + 2[S(IV)]\right) \tag{55}$$

$$A_2 = 1 - \frac{K_w}{K_{su}K_{bi}} - \frac{[S(IV)]}{K_{bi}} \tag{56}$$

$$A_3 = \frac{1}{K_{bi}} \tag{57}$$

$$A_4 = \frac{1}{K_{su}K_{bi}} \tag{58}$$

$[S(IV)]$ is the concentration of dissolved total tetravalent sulfur according to Henry's law.

Henry's law is also used to calculate the initial concentration of dissolved oxygen. The initial concentrations of Fe^{2+} and $SO_4{}^{2-}$ are zero, and the initial concentration of OH^- is calculated according to the previously obtained H^+ concentration.

The setting of the boundary conditions is an important point for this model because it helps to determine the corrosion reaction in charge transfer control or diffusion control state. The gradient of all components at the right boundary should be zero because the dissolution of the corrosion products and ionizing components in supercritical CO_2 is ignored, except for SO_2 and O_2

$$\left.\frac{\partial c_j}{\partial x}\right|_{x=d_i} = 0 \tag{59}$$

For O_2, the right boundary condition is that the concentration is equal to the initial value obtained through Henry's law

$$\left. c_{O_2}\right|_{x=d_i} = c^0_{O_2} \tag{60}$$

The total concentration of tetravalent S is equal to the value calculated by Henry's law. Therefore, the right boundary condition of SO_2 is

$$\left.[SO_2 \cdot H_2O]\right|_{x=d_i} = [S(IV)] - [HSO_3^-] - [SO_4^{2-}] \tag{61}$$

The situation is relatively complicated for the left boundary not only because of the occurrence of electrochemical reactions and the consumption of reactants but also because of the occurrences

Table 1
List of model parameter and constant inputs.

Symbol	Value	Unit	Definition	References
Constants				
F	96485.3399	C/mol	Faraday constant	[55]
R_g	8.31451	J/(mol K)	Molar gas constant	[56]
Physical				
d_h	0.1265	m	Hydraulic diameter	Measured
Sulfite oxidation				
A_k	9.17		Pre-exponential factor	[57]
ΔH	25,650	J/mol	Activation energy	[57]
Electrochemistry				
T_{ref}	298	K	Electrode reference temperature	[7]
α_a	1.5		Anode transfer coefficient	[5]
α_c	0.5		Cathode transfer coefficient	[5]
$E_{Fe,rev}$	−0.488	V	Anode equilibrium potential	[53]
i_{0,H^+}^{ref}	0.05	A/m^2	Reference exchange current density for hydrogen electrode	[5]
$i_{0,Fe}^{ref}$	1	A/m^2	Anode reference exchange current density	[53]
ΔH_{H^+}	30,000	J/mol	Activation energy of hydrogen electrode exchange current density	[5]
ΔH_{Fe}	37,500	J/mol	Activation energy of anode exchange current density	[53]
Product film				
ρ_{FeSO_3}	2096	kg/m^3	Density of $FeSO_3 \cdot 3H_2O$	Experiment value
ρ_{FeSO_4}	2290	kg/m^3	Density of $FeSO_4 \cdot 4H_2O$	[58]
Activity coefficients				
b_1	1.9897		Modified TCPC model parameter	[46]
S_1	3.3404		Modified TCPC model parameter	[46]
n_1	0.9473		Modified TCPC model parameter	[46]
Precipitation				
k_G	0.218		Effective crystal growth rate constant	[51]
n_s	1.6		Exponent	[51]
Molecular diffusion coefficients				
D_{m,H^+}	9.311×10^{-9}	m^2/s	H^+ diffusion coefficient	[55]
D_{m,HSO_3^-}	1.545×10^{-9}	m^2/s	$HSO_3{}^-$ diffusion coefficient	[55]
$D_{m,SO_3^{2-}}$	0.959×10^{-9}	m^2/s	$SO_3{}^{2-}$ diffusion coefficient	[55]
D_{m,SO_2}	1.862×10^{-9}	m^2/s	SO_2 diffusion coefficient	[59]
D_{m,O_2}	1.96×10^{-9}	m^2/s	O_2 diffusion coefficient	[60]
$D_{m,Fe^{2+}}$	0.719×10^{-9}	m^2/s	Fe^{2+} diffusion coefficient	[55]
$D_{m,SO_4^{2-}}$	1.065×10^{-9}	m^2/s	$SO_4{}^{2-}$ diffusion coefficient	[55]

of precipitation reactions, which lead to changes in the relevant component concentrations. A comparison of SO_2 and O_2 fluxes at the left boundary is needed. For a smaller flux component, its left boundary condition is

$$c_s|_{x=0} = 0 \tag{62}$$

c_s represents the smaller flux component at the left boundary. For the larger flux component, its left boundary condition is

$$N_b\big|_{x=0} = N_s\big|_{x=0} \tag{63}$$

N_b and N_s represent the flux of the larger flux component and the smaller flux component at the left boundary, respectively.

For H^+ ion, the left boundary condition at the charge-transfer control stage is

$$N_{H^+}\big|_{x=0} = -\frac{i_{H^+}}{F} \tag{64}$$

The left boundary condition at the diffusion control stage is

$$c_{H^+}\big|_{x=0} \approx 0 \tag{65}$$

In the actual calculation, H^+ would not completely equal zero, but it would stabilize at a low value.

For $HSO_3{}^-$ ion, its left boundary condition is

$$\left.\frac{\partial c_{HSO_3^-}}{\partial x}\right|_{x=0} = 0 \tag{66}$$

For $SO_3{}^{2-}$ ion, due to the precipitation reaction, its left boundary condition is

$$N_{SO_3^{2-}}\Big|_{x=0} = R_{G1} \tag{67}$$

For $SO_4{}^{2-}$ ion, consumption of precipitation and generation from oxygen electrode reaction exist, so its left boundary condition is

$$N_{SO_4^{2-}}\Big|_{x=0} = R_{G2} - \frac{i_{O_2}}{2F} \tag{68}$$

For Fe^{2+} ion, its left boundary condition is

$$N_{Fe^{2+}}\big|_{x=0} = R_{G1} + R_{G2} - \frac{i_{Fe^{2+}}}{2F} \tag{69}$$

3.2. Numerical method

The model solves a problem with variable grids because of corrosion product film thickening. The central explicit difference method is used to solve these partial differential equations. The model includes eight components wherein the distribution of OH^- ions can be directly obtained by H^+ distribution. The distribution of the other seven components requires solving the partial differential equations simultaneously because the components are coupled in the reactions. The discrete form of Eq. (16) is as follows:

$$\varepsilon_m \frac{c_{j,m}^{n+1} - c_{j,m}^{n}}{\tau} = -\varepsilon_m^{1.5} D_{j,m} \frac{c_{j,m+1}^{n} - 2c_{j,m}^{n} + c_{j,m-1}^{n}}{h^2} + \varepsilon_m R_{j,m}^{n} \left(CR^n - \frac{d_f^{n+1} - d_f^{n}}{\tau} \right) \frac{c_{j,m+1}^{n} - c_{j,m-1}^{n}}{2h} \tag{70}$$

The model parameter and constant inputs are shown in Table 1. Similar to Nordsveen's model [7], the concentrations of all components are stored in a matrix. This method is also used to treat diffusion coefficients and other parameters.

Table 2
Data of predicted corrosion rates and measured corrosion rates (experimental data taken from Dugstad et al. [18], Choi et al. [17], Farelas et al. [25], and Xiang et al. [20,22–24].).

No.	Condition							Predicted CR (mm/a)	Measured CR (mm/a) and Ref.
	T (K)	P (MPa)	mp_{SO_2}	mp_{O_2}	RH	U (rpm)	t (h)		
1	293.15	10.0	0.069	0.014	17	0	168	0.0012	0.01 [18]
2	323.15	8.0	1.0	0	100	0	24	1.1	5.6 [17]
3	323.15	8.0	1.0	4.125	100	0	24	6.99	7 [17]
4	298.15	8.0	0.1	0	22	0	24	0.00075	0.1 [25]
5	298.15	8.0	0.05	0	22	0	24	0.000259	~0 [25]
6	323.15	8.0	1.0	0	19	0	24	0.0173	3.5 [25]
7	323.15	8.0	0.1	0	19	0	24	0.000383	0.03 [25]
8	323.15	8.0	0.05	0	19	0	24	0.000137	0.05 [25]
9	323.15	10	0.2	0.11	100	120	288	0.714	0.193, 0.339 [20]
10	323.15	10	0.7	0.11	100	120	288	0.939	0.678, 1.351 [20]
11	323.15	10	1.4	0.11	100	120	288	1.138	0.847, 1.129 [20]
12	323.15	10	2.0	0.11	100	120	288	1.377	0.876, 0.962 [20]
13	323.15	10	2.0	0.11	9	120	120	0.0154	0.00491 [22]
14	323.15	10	2.0	0.11	50	120	120	0.0853	0.0387 [22]
15	323.15	10	2.0	0.11	60	120	120	0.101	0.0775 [22]
16	323.15	10	2.0	0.11	70	120	120	0.542	0.33 [22]
17	323.15	10	2.0	0.11	88	120	120	1.317	0.862 [22]
18	323.15	10	2.0	0.11	100	120	120	1.825	1.457 [22]
19	323.15	10	2.0	0.11	100	120	24	3.061	1.946 [24]
20	323.15	10	2.0	0.11	100	120	72	2.119	1.773 [24]
21	323.15	10	2.0	0.11	100	120	192	1.58	0.728 [24]
22	298.15	10	2.0	0.06	100	120	120	1.074	1.096 [23]
23	323.15	10	2.0	0.11	100	120	120	1.788	2.53 [23]
24	348.15	10	2.0	0.18	100	120	120	3.432	3.06 [23]
25	366.15	10	2.0	0.22	100	120	120	1.271	1.256 [23]

Note: For nos. 9–12, the latter experimental values are for the iron samples.

This model uses the non-uniform space interval grid to increase the time step. The time step is on the order of 10^{-2} s. The grid number is generally between 20 and 30 because too many grids will greatly increase the required computing time.

This model uses the chemical equilibrium assumption wherein continuous calculation of chemical equilibrium is needed to constantly solve the following nonlinear equations:

$$\frac{(c_1+x)(c_2+y)}{c_4+w} = K_{su} \tag{71}$$

$$\frac{(c_1+x)(c_3+z)}{c_2+y} = K_{bi} \tag{72}$$

$$x = y + 2z \tag{73}$$

$$w + y + z = 0 \tag{74}$$

where c_1, c_2, c_3, and c_4 represent the concentrations of H^+, $HSO_3{}^-$, $SO_3{}^{2-}$, and SO_2 at non-equilibrium state, respectively. x, y, z, and w represent the variation from a non-equilibrium state to an equilibrium state. The following approximate calculation methods were used to solve these equations:

If $c_1c_2 < c_4K_{su}$, then H^+ and $HSO_3{}^-$ are generated, and the amount is

$$x_1 = \frac{-(K_{su}+c_1+c_2)+\sqrt{(K_{su}+c_1+c_2)^2-4(c_1c_2-K_{su}c_4)}}{2} \tag{75}$$

If $c_1c_3 < c_2K_{bi}$, then H^+ and $SO_3{}^{2-}$ are generated, and the amount is

$$x_2 = \frac{-(K_{bi}+c_1+c_3)+\sqrt{(K_{bi}+c_1+c_3)^2-4(c_1c_3-K_{bi}c_2)}}{2} \tag{76}$$

However, a thin layer adjacent the substrate is treated as the non-equilibrium state.

Due to lack of relevant data for $FeSO_3$ and $FeSO_4$ in the modified TCPC model [46], this model used the data of $MnSO_4$ instead.

4. Results and discussion

4.1. Model verification

The predicted corrosion rates and measured corrosion rates from the open literature were compared to validate the model reliability, with the conditions listed in Table 2. Thus far, published data concerning the corrosion rate of pipeline steel corrosion in supercritical CO_2–SO_2–O_2–H_2O are limited. Most of the available data are presented in graphs, which makes experimental data difficult to obtain. Thus, some of the experimental data in Table 2 were obtained through graph measurements.

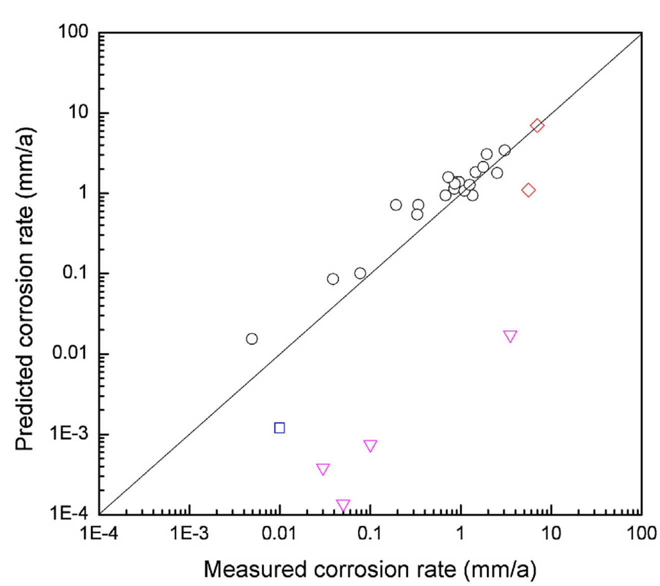

Fig. 2. Comparison between measured corrosion rates and predicted corrosion rates (▽ Dugstad et al. [18]; ◊ Choi et al. [17]; ○ Xiang et al. [20,22–24]; △ Farelas et al. [25]).

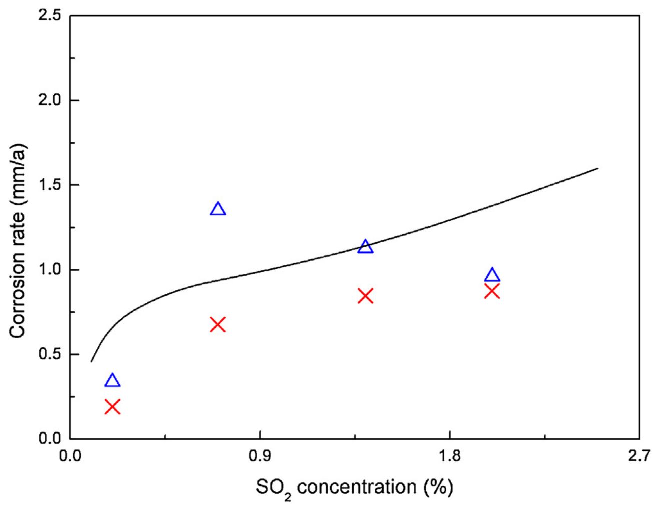

Fig. 3. Predicted corrosion rates and experimental results with the variation of SO_2 concentration (— Predicted value; △ Measured value of iron sample [20]; × Measured value of X70 steel sample [20]).

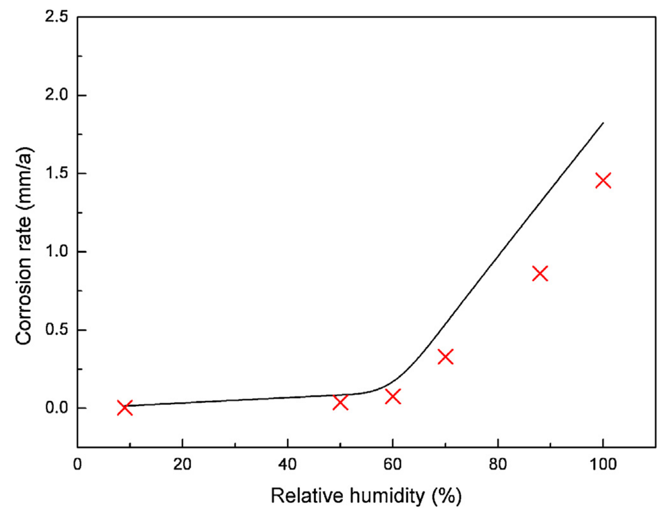

Fig. 4. Predicted corrosion rates and experimental values with the variation of relative humidity (— predicted value; × measured value [22]).

The predicted and measured results are also shown in Fig. 2. The comparison between predicted and measured corrosion rates shows that this model can provide satisfactory prediction results, especially in cases with high SO_2 concentration and relative humidity. When SO_2 concentration was extremely low, the model seemed unable to accurately predict experimental results, possibly because when the SO_2 concentration is extremely low, the effect of dissolved CO_2 should be taken into consideration. Under unsaturated humidity condition, the corrosion mechanisms also need further understanding.

4.2. Effect of SO_2 concentration

Fig. 3 shows the predicted corrosion rates and experimental results [20] for X70 steel and iron with variation of SO_2 concentration. The model results show that as the SO_2 concentration increases, the corrosion rate also increases. The model also shows a faster increasing trend when the SO_2 concentration is low. Except for the case of 0.7% SO_2 concentration where the localized corrosion occurred at the bottom of the iron sample [20], the model results overpredict the experimental results. During the experiment, the components were gradually consumed because the autoclave was a closed system. Therefore, at the end of the experiment, the SO_2 concentration was lower than the initial concentration, whereas the SO_2 concentration was assumed to be invariant in model calculation.

4.3. Effect of relative humidity

Fig. 4 presents the predicted corrosion rates and experimental values of X70 steel with the varization of relative humidity [22]. As observed, the model results are larger than the experimental values. During the experiment, the relative humidity was gradually reduced in the closed autoclave environment. The presence of SO_2 likely increased the solubility of water in CO_2, whereas the relative humidity was calculated in accordance to the case of pure CO_2. Austegard et al. [61] showed that the presence of CH_4 increased the water solubility in supercritical CO_2. However, the effect of SO_2 was not found in the related literature. These reasons all make it difficult to give accurate modeling.

4.4. Effect of time

Fig. 5 shows the predicted corrosion rates and experimental values [20,24] for X70 steel with the variation of time. The predicted corrosion rates are the average corrosion rates of the entire period. The model results show that the predicted corrosion rate is high during the initial stage. For example, when corrosion time is 1 h, the corrosion rate reaches 26 mm/a, in agreement with the experimental result obtained by electrical resistance method with a corrosion rate of 56 mm/a at the initial stage [20]. With prolonged time, the predicted corrosion rate decreases sharply. When the time is longer than 24 h, the corrosion rates decrease slowly. The model calculation results show that the initial pH value is generally less than two. However, as the corrosion reaction proceeds, the pH value of liquid film increases because of the accumulation of HSO_3^-, and the corrosion product film is gradually formed, if the water refreshing rate is low for the water film. The formation of corrosion product film also inhibits the corrosion significantly.

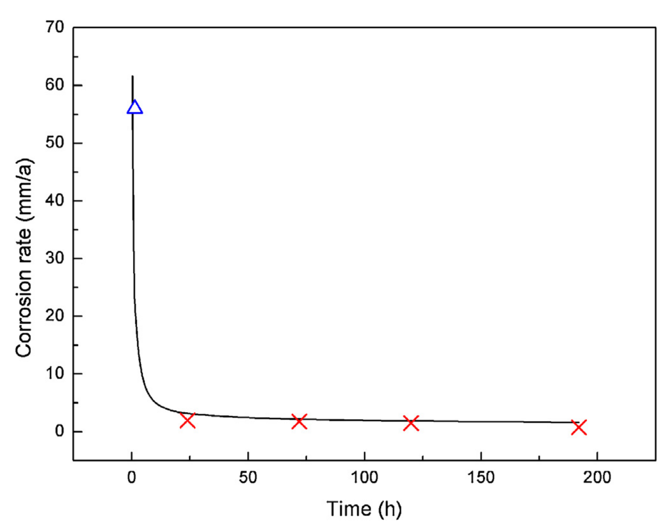

Fig. 5. Predicted corrosion rates and experimental results with the variation of time (— predicted value; △ measured value of iron foil sample [20]; × measured value of X70 steel sample [24]).

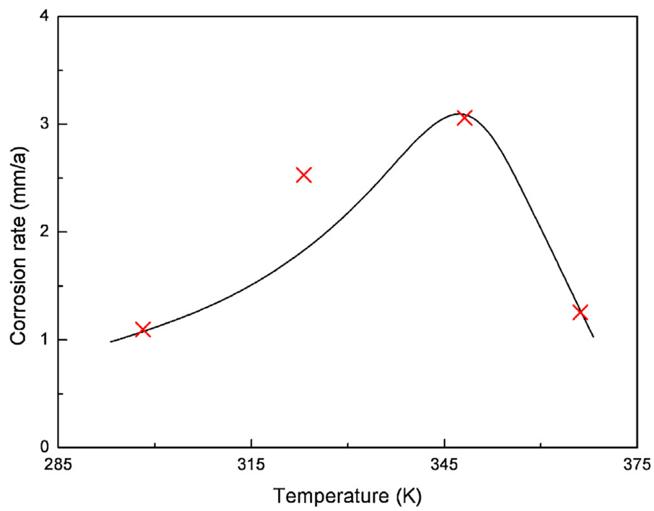

Fig. 6. Measured corrosion rates and predicted results with the variation of temperature (— predicted value; × measured value [23]).

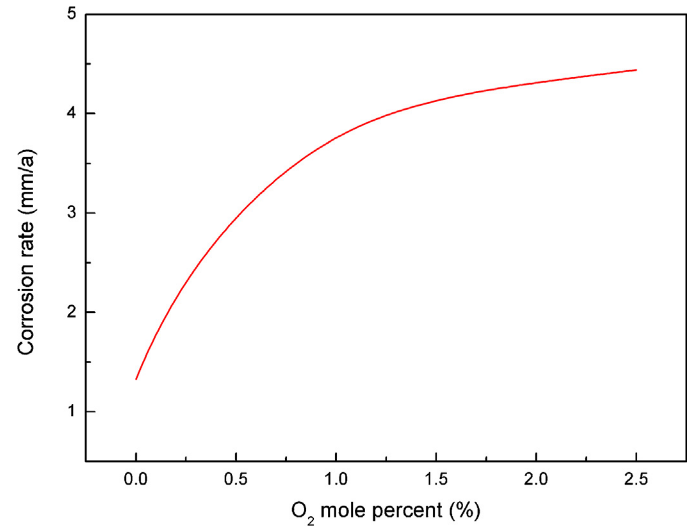

Fig. 8. Predicted variation of corrosion rate with O_2 mole fraction (323.15 K, 10 MPa, 2.0% SO_2, 0–2.5% O_2, 100% RH, 120 rpm, 120 h).

4.5. Effect of temperature

Fig. 6 shows the measured corrosion rates [23] and predicted values with the varization of temperature. The model can predict the trend of corrosion rate changing with temperature. As shown in Fig. 6, the corrosion rate first increases and then decreases. Under high-temperature conditions, the corrosion product film is dense, which effectively decreases the corrosion rate. Under low temperature, the diffusion coefficients of components and the chemical reaction rates are low; therefore, the corrosion rate is low.

4.6. Effect of other factors

Thus far, the effects of pressure, O_2, and velocity on the corrosion rate in supercritical CO_2–SO_2–O_2–H_2O environments are still unclear. No experimental data can be used to compare with the present model prediction results. This model is used to investigate the effects of pressure, O_2 concentration, and velocity on the corrosion rates.

Fig. 7 shows the effect of total CO_2 mixture pressure on the corrosion rate calculated by the current model. As the pressure increases, the corrosion rate presents a monotonically increasing trend. The experimental results of Choi and Nesic [62] showed that when the pressure is increased from 6 MPa to 8 MPa, the corrosion rate increases in supercritical CO_2. In the calculation model, the increase in pressure indicates that under the conditions with the same SO_2 and O_2 mole fraction, these two components have higher partial pressures and their dissolved amount in water film increase, thereby increasing the corrosion rate proportionally.

Fig. 8 presents the changes of corrosion rate with O_2 concentration obtained by the model calculation. As O_2 concentration increases, the corrosion rate shows a gradually increasing trend. The rate of increase slows down gradually. The experimental results of Dugstad et al. [18] proved that the existence of O_2 accelerated the corrosion process, which is consistent with the results of the present numerical experiments.

Fig. 9 shows the changes of corrosion rate with autoclave rotation speed. The corrosion rates increase in direct proportion with the rotation speed. When the rotation speed is relatively low, the corrosion rate shows a gradually increasing trend with the rotation speed. When the rotation speed is faster than 720 rpm, the corrosion rate remarkably increases. The experimental results of Dugstad et al. [18] showed that an increase in velocity sharply increases the corrosion rate.

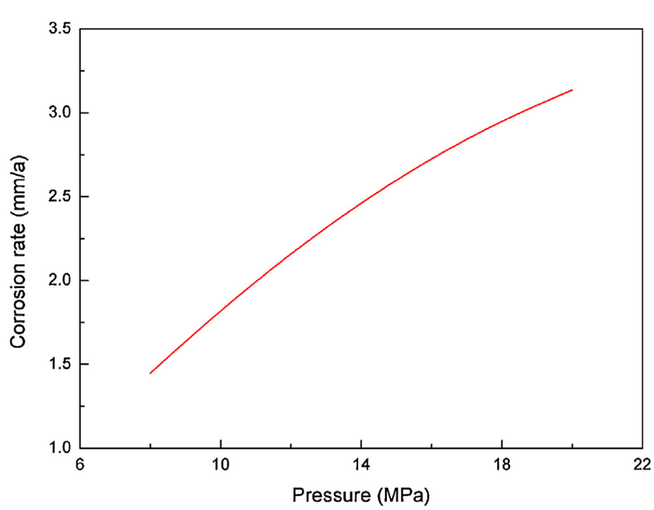

Fig. 7. Predicted variation of corrosion rate with pressure (323.15 K, 8–20 MPa, 2.0% SO_2, 0.1% O_2, 100% RH, 120 rpm, 120 h).

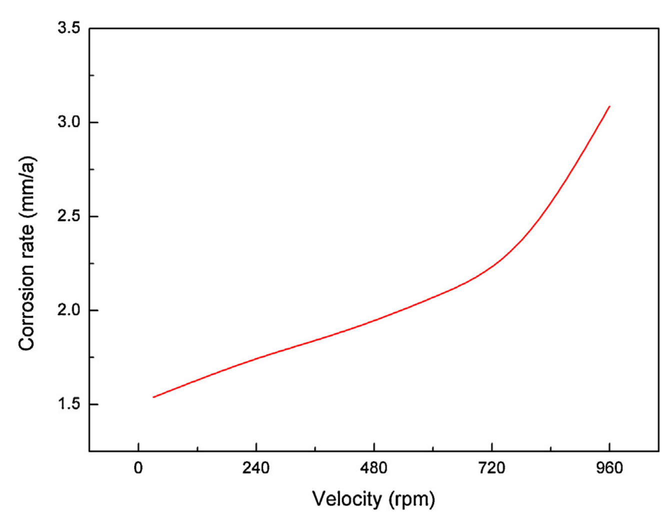

Fig. 9. Predicted variation of corrosion rate with velocity (323.15 K, 10 MPa, 2.0% SO_2, 0.1% O_2, 100% RH, 30–960 rpm, 120 h).

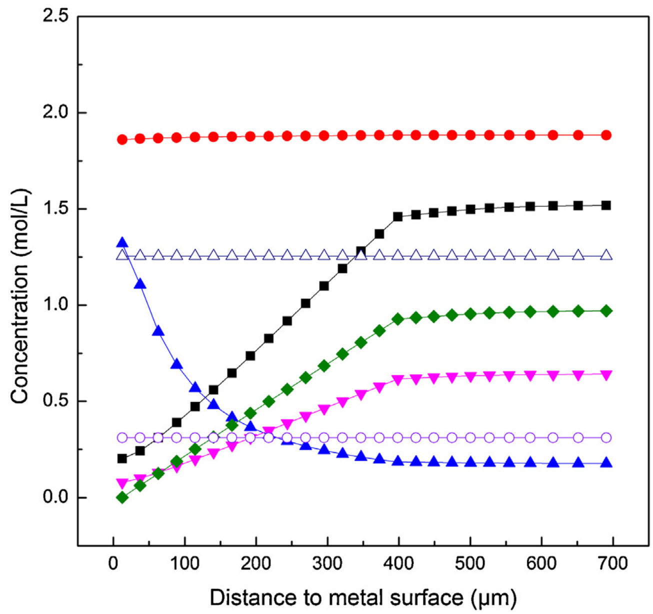

Fig. 10. Component concentration distributions in water film and porous product film (323.15 K, 10 MPa, 2.0% SO_2, 0.1% O_2, 100% RH, 120 rpm, 120 h. (—●—) HSO_3^-; (—■—) 1000×H^+; (—△—) Fe^{2+}; (—◆—) 15,000×O_2; (—▼—) 10×$SO_2{\cdot}H_2O$; (—○—) SO_4^{2-}; (—▲—) 100×SO_3^{2-};.).

4.7. Concentration distributions

Fig. 10 presents the component concentration distributions in the product and water film. The figure shows that greater concentration gradients exist in the product film than in the water film. Ignoring the impact of the ion activity, the pH value at the outside surface of the liquid film is about 2.8, and the pH value is approximately 3.7 near the metal substrate. The concentrations of HSO_3^-, SO_4^{2-}, and Fe^{2+} remain approximately stable.

The calculated ionic strength of the water film is about 4.1 mol/kg under this condition. In some cases, the ionic strength of the water film even reaches about 10.0 mol/kg. These data show that the water film in the current research environment is a high-ionic strength environment, showing that the introduction of modified TCPC model to calculate the activity coefficient is necessary.

5. Conclusions

Based on the CO_2 corrosion models and the atmospheric corrosion model, this study established a mechanistic model suitable for pipeline corrosion in supercritical CO_2–SO_2–O_2–H_2O environments by dividing the physical area into six regions (SIWDES) and introducing the modified TCPC model to calculate the activity coefficients of the components in high-ionic strength solution. The model reasonably reflected the effect of primary variables on corrosion rate. The model also showed the concentration distributions of components in the water and product films. A comparative analysis between the model and experimental results showed that the model provided the correct influential trends of variables on the corrosion rates. This model revealed the mechanisms by which impact factors affect the corrosion rates of pipeline steel corrosion in supercritical CO_2–SO_2–O_2–H_2O environments, which had the values of academic and industrial applications.

Acknowledgements

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Appendix A. Parameters with temperature

Stokes–Einstein equation is used to describe the relationship of component diffusion coefficients with temperature [55]:

$$D_m = D_{m,ref}\frac{T}{T_{ref}}\frac{\mu_{ref}}{\mu} \quad (A1)$$

The variable relationship of viscosity and temperature is [55]

$$\mu = \mu_{ref} \times 10^{\sigma} \quad (A2)$$

$$\sigma = \frac{1.3277 \times (293.15 - T) - 0.001053 \times (293.15 - T)^2}{T - 168.15} \quad (A3)$$

The reaction rate constants with respect to temperature follow

$$k(T) = A_k \exp\left(\frac{-\Delta H}{RT}\right) \quad (A4)$$

This equation can be further written as

$$k = k_{ref} \exp\left[\frac{\Delta H}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right] \quad (A5)$$

The Henry's constant H_{SO_2} for SO_2 dissolved in water follows [63]

1.7   7   0.40   9.1 −4
6

$$\ln\ H_{SO_2} = \frac{(510/T_0 - 26,970T_1 + 155T_2 - 0.0175T_0T_3)}{R_g} \quad (A6)$$

The first ionization equilibrium constant of SO_2 hydrate with respect to temperature follows

$$\ln\ K_{su} = \frac{(-10600/T_0 - 17,800T_1 - 272T_2 - 0.85T_0T_3)}{R_g} \quad (A7)$$

The second ionization equilibrium constant of SO_2 hydrate with respect to temperature follows

$$\ln\ K_{bi} = \frac{(-40,940/T_0 - 3650T_1 - 262T_2 - 1.35T_0T_3)}{R_g} \quad (A8)$$

where

$$T_0 = 298.15\ \ K \quad (A9)$$

$$T_1 = \frac{1}{T_0} - \frac{1}{T} \quad (A10)$$

$$T_2 = \frac{T_0}{T} - 1 + \ln\left(\frac{T}{T_0}\right) \quad (A11)$$

$$T_3 = \frac{T}{T_0} - \frac{T_0}{T} - 2\ \ln\left(\frac{T}{T_0}\right) \quad (A12)$$

For the three equilibrium constants of SO_2, the model uses the changed forms of the above relationship:

4. e40

$$\ln\ H_{SO_2} = \frac{-142.679 + 8988.76}{T} + 19.8967\ \ln\ T - 0.0021T \quad (A13)$$

$$\ln\ K_{su} = \frac{554.963 - 16700.1}{T} - 93.67\ \ln\ T + 0.1022T \quad (A14)$$

$$\ln\ K_{bi} = \frac{-358.57 + 5477.1}{T} + 65.31\ \ln\ T - 0.1624T \quad (A15)$$

The relationship between Henry's constant and temperature is [64]

$$\boxed{H(T) = H_0 \exp\left[C\left(\frac{1}{T} - \frac{1}{298}\right)\right]} \quad (A16)$$

The Henry's constant of oxygen dissolved in water with respect to temperature follows [48]

$$\ln\ H_{O_2} = \frac{0.298399 - 5596.17}{T} + \frac{1,049,668}{T^2} \tag{A17}$$

Fitted from the experimental data of Harned et al. [65], the water ionization constant with respect to temperature follows

$$-\log\ K_w = 55.17462 - 0.29688T + 7.132 \times 10^{-4}T^2 - 6.06061 \times 10^{-7}T^3 \tag{A18}$$

For the solubility of $FeSO_4$, based on the data of Sohnel and Novotny [66], the model uses piecewise linear fitting to describe its relationship with temperature:

$$S_{FeSO_4} = \begin{cases} -89.622 + 0.37723T & T < 328.15 \\ -88.049 + 0.96625T + 0.00179T^2 & T \geq 328.15 \end{cases} \tag{A19}$$

Relevant experimental data for the solubility of $FeSO_3$ are very limited. The solubility of $FeSO_3$ in pure water is very low [67]. According to the experimental data of Terres and Ruhl [68], the following formula is used to describe the relationship between $FeSO_3$ solubility and SO_2 concentration in the model:

$$S_{FeSO_3} = 100 \times \frac{0.03058c_{SO_2} + 0.02630}{0.03058c_{SO_2} + 0.02630 + 1} \tag{A20}$$

Based on the data of Harr et al. [69–71], the influence of temperature on the density of water is

$$\rho_w = 753.596 + 1.87747T - 0.00356T^2 \tag{A21}$$

Based on the data of Archer and Wang [72], the relationship between the dielectric constant of water and temperature is

$$D_e = 250.613 - 0.79541T + 7.30653 \times 10^{-4}T^2 \tag{A22}$$

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